

Modelling and simulation of affinity membrane adsorption

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Abstract

A mathematical model for the adsorption of biomolecules on affinity membranes is presented. The model considers convection, diffusion and adsorption kinetics on the membrane module as well as the influence of dead end volumes and lag times; an analysis of flow distribution on the whole system is also included. The parameters used in the simulations were obtained from equilibrium and dynamic experimental data measured for the adsorption of human IgG on A2P-Sartoeoxy affinity membranes. The identification of a bi-Langmuir kinetic mechanisms for the experimental system investigated was paramount for a correct process description and the simulated breakthrough curves were in good agreement with the experimental data. The proposed model provides a new insight into the phenomena involved in the adsorption on affinity membranes and it is a valuable tool to assess the use of membrane adsorbers in large scale processes.

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1. Introduction

High purity products are often required from the biotechnology industry; in order to achieve the demanded purity, several complex purification steps are needed [1]. Among them, affinity chromatography is the preferred choice for the primary capture step, in general and in particular in monoclonal antibody production [2]. This purification method is based on the specific interaction between the biomolecule of interest and a ligand immobilized on a solid support.

Typically, the process is carried out in bead-based packed columns. Several major limitations affect this *modus operandi*, such as high pressure drops, compressibility of the beads and slow mass transfer [3]. To overcome these problems, a new approach based on microporous membranes as chromatographic media has been developed in the last decades. Membrane affinity chromatography can be operated at higher flow rates, since pressure drops in a layered-stack are much smaller than in packed columns and mass transport occurs mainly by convection. As a consequence, faster processes are possible, implying higher productivity and lower possibility of protein denaturation.

The analysis and modelling of protein adsorption on affinity membranes has been investigated by many research groups.

The basic approach for single protein adsorption, derived from frontal analysis of chromatographic systems, is the combination of the continuity equation coupled with a kinetic equation that describes the protein adsorption mechanism, and different models have been developed depending on the geometry considered. Briefs and Kula analyzed a single cylindrical pore and then extended the solution over the whole membrane [4], Suen and Etzel considered a circular flat sheet membrane with cylindrical pores [5], Liu and Fried studied the influence of pore size distribution and membrane thickness [6]. A more general approach was offered by Serafica et al. who examined a hollow fibre geometry, whose results can be easily extended to the case of a flat sheet system [7]. In the latter work the agreement with experimental data was very good at the onset of breakthrough, but a mismatch was observed at later stages of breakthrough due to asymmetry of the curves that cannot be described by the Langmuir model.

Broadening of breakthrough curves near the saturation of the membrane layers has been observed [8–10] and it gave the indication that a different kinetic mechanism needs to be considered.

In the steric hindrance model, also known as the “car parking model” [11], the adsorption becomes dependent on the availability of binding sites, which allows to consider decreased accessibility of the active sites to the protein when surface coverage increases. The steric hindrance model has been implemented in the mathematical description of membrane adsorbers by Yang

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and Etzel [10]. However, the comparison with experimental data was not highly satisfactory, indicating that steric hindrance was not the dominating mechanism.

A different theory, presented as the spreading model, considers the conformational changes that a protein may experience when bound to a surface, during the process leading to the minimal free energy configuration [12]. By using the spreading model Yang obtained a very good fitting of the breakthrough curves [10], nevertheless the implementation of the spreading model increases the number of fitting parameters required for the solution and an extensive experimentation is envisaged for model validation.

Shi et al. used the Freundlich model to describe the kinetics of adsorption. The agreement of the simulated breakthrough curves with the experimental data was very good in the first part of the curve, but at saturation a discrepancy is still present [13].

Wang and Carbonell observed that the adsorption of a target protein onto a solid support can involve independent binding sites associated with different energies [14]; they postulate the existence of high energy and low energy binding sites. Furthermore, from a general point of view, all sites could be characterized by a different kinetic mechanism and a different reaction energy.

In the present work, IgG adsorption on A2P-Sartoepoxy membranes was described by a bi-Langmuir kinetics as it is suggested by several experimental observations, including the high non-specific adsorption [15] and double elution peak [16]. Moreover, flow non-idealities and delay time, due to the external volumes of the system, have been described by a simple system dispersion model [17] that has been included in the comprehensive mathematical formulation presented.

The model has been used to simulate the breakthrough curves of IgG on A2P-Sartoepoxy membranes using equilibrium parameters derived from batch experiments. The results obtained and the relevant fitting parameters have been analyzed and discussed in detail.

2. Theory

2.1. Equilibrium adsorption isotherms

The Langmuir model is widely used to describe the adsorption of a single protein on a solid support. This model considers only monovalent binding under the assumption of homogeneous adsorption, namely the surface is covered by a monolayer with no interactions between adsorbed species and the free active sites [5,10,18]. The reaction between the protein P and the ligand L can be schematized as:



where PL represents the complex protein–ligand. At equilibrium, the Langmuir isotherm can be written in the following form:

$$c_s = \frac{c_1 c}{K_{d,1} + c} \quad (2)$$

in which c_s is the concentration of the protein–ligand complex adsorbed on the membrane, c_1 is the static binding capacity of the membrane, c is the protein concentration in the liquid phase and K_d is the Langmuir dissociation constant.

However, when the adsorption involves different binding sites, associated to different energies, a protein can bind to more than one adsorption site with different mechanisms. Since the affinity membranes used in this work, A2P-Sartoepoxy, have shown a high non-specific binding in a series of control experiments, a bi-Langmuir model has been chosen to better describe the system [14]. The corresponding isotherm can be written in the following form:

$$c_s = \frac{c_{1,1} c}{K_{d,1} + c} + \frac{c_{1,2} c}{K_{d,2} + c} \quad (3)$$

in which subscript 1 labels the contribution of specific binding and subscript 2 the contribution due to non-specific binding.

The experimental equilibrium data have been described by best-fitting Eq. (3) to the experimental data using three adjustable parameters, namely the two Langmuir dissociation constants, $K_{d,1}$ and $K_{d,2}$, and the membrane capacity for the specific adsorption, $c_{1,1}$. The static binding capacity associated to non-specific binding has been taken from experimental data obtained in independent control experiments with HSA [15]. The best-fitting parameters were obtained through a minimization of the square residual error between the experimental data and Eq. (3).

2.2. System dispersion

In order to give an appropriate description of the affinity membrane adsorption kinetics, the entire system must be considered to account for the flow non-idealities such as dispersion, mixing, channelling and dead volumes [17,19]. All these phenomena are often globally indicated as system dispersion. A system dispersion model has been used to describe such effects over the entire flow system used experimentally. To that aim we considered the system volume V_{sys} , which is the sum of the membrane void volume, V_{void} , and the external volume, V_{ext} , which comprises the volume of flow distributors, the volume of the detector flow cell, the volume of the pump head, etc. In symbols:

$$V_{sys} = V_{void} + V_{ext} \quad (4)$$

A combination in series of an ideal continuously stirred tank reactor, CSTR, and of an ideal plug flow reactor, PFR, has been used to model the flow behaviour of the system as illustrated in Fig. 1. As usual, the CSTR model describes the broadening of the step response, whilst the PFR model describes the time delay in the step response. In this way the system volume can be expressed also as:

$$V_{sys} = V_{CSTR} + V_{PFR} \quad (5)$$

The classical CSTR model mass balance gives:

$$\frac{dc_{out}^{CSTR}}{dt} = \frac{Q}{V_{CSTR}} (c_{in}^{CSTR} - c_{out}^{CSTR}) \quad (6)$$

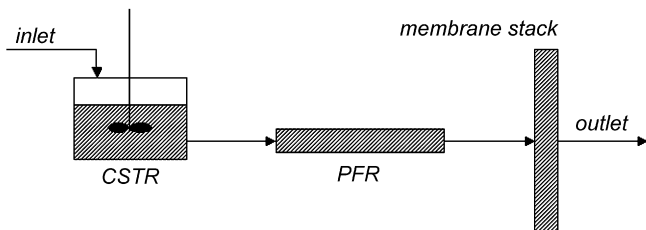


Fig. 1. Diagram of system dispersion model for layered-stack device.

$$c_{\text{out}}^{\text{CSTR}} = 0 \quad \text{at} \quad t = 0 \quad (7)$$

where Q indicates the volumetric flow rate, $c_{\text{in}}^{\text{CSTR}}$ and $c_{\text{out}}^{\text{CSTR}}$ represent the inlet and outlet concentrations, respectively. The PFR is represented by a pure time delay, i.e.:

$$c_{\text{out}}^{\text{PFR}}(t) = \begin{cases} 0 & t < t_d \\ c_{\text{in}}^{\text{PFR}} & t \geq t_d \end{cases} \quad (8)$$

where the delay time t_d is equal to V_{PFR}/Q .

Eqs. (6)–(8) have been written in dimensionless form:

$$\frac{dC_{\text{out}}^{\text{CSTR}}}{d\tau} = \frac{1}{\tau'_H} (C_{\text{in}}^{\text{CSTR}} - C_{\text{out}}^{\text{CSTR}}) \quad (9)$$

$$C_{\text{out}}^{\text{CSTR}} = 0 \quad \text{at} \quad \tau = 0 \quad (10)$$

$$C_{\text{out}}^{\text{PFR}}(\tau) = \begin{cases} 0 & \tau < \tau_d \\ C_{\text{in}}^{\text{PFR}} & \tau \geq \tau_d \end{cases} \quad (11)$$

where C_j are dimensionless concentrations, τ_d is the dimensionless PFR time delay and τ'_H is a time constant, respectively defined as follows:

$$C_j = \frac{c_j}{c_0} \quad (12)$$

$$\tau_d = \frac{V_{\text{PFR}}}{Q} \frac{v}{L} \quad (13)$$

$$\tau'_H = \frac{V_{\text{CSTR}}}{Q} \frac{v}{L} \quad (14)$$

According to Fig. 1, the response of the CSTR is used as the inlet concentration for the PFR model, and the PFR exit concentration is used as inlet concentration for the dynamic model of the membrane system. Furthermore, since the inlet concentration to the CSTR is c_0 , the set of Eqs. (9)–(11) is solved analytically and the inlet concentration to the membrane stack can thus be easily expressed in the following dimensionless form:

$$C_{\text{in}}(\tau) = \begin{cases} 0 & \tau < \tau_d \\ 1 - e^{(-1/\tau'_H)(\tau - \tau_d)} & \tau \geq \tau_d \end{cases} \quad (15)$$

In an ideal system, the breakthrough curve of a non-binding solute is a vertical line, while in a real system it has the classical sigmoidal shape and it is generally indicated as system dispersion curve. The system dispersion curve characterizes the dispersive flow behaviour of the experimental system used, independently of the binding mechanism between ligand and ligate. System dispersion curves are extremely important in laboratory

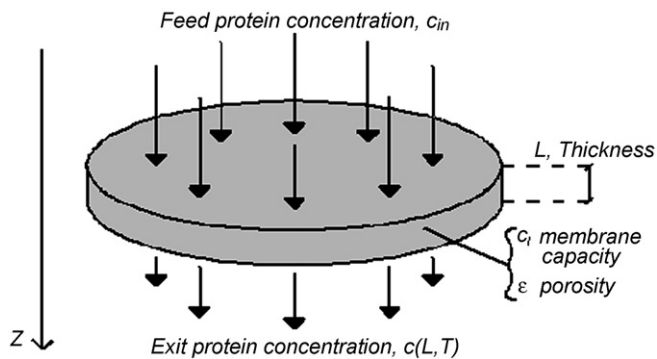


Fig. 2. Schematic drawing of an affinity membrane.

scale membrane units, since the external volumes are often of the same order of magnitude of the membrane volume [20], as in the case under investigation. These curves can be obtained through direct experimental measurements by feeding a non-adsorbing solute to the membranes, and are then used to evaluate the actual inlet concentration in the membrane stack as a function of time.

2.3. Modelling dynamic adsorption on the membrane

The model is based on the isothermal sorption of a single protein when the pure protein solution flows through a porous membrane as schematized in Fig. 2. Viscous flows are considered in all cases in view of the rather small Reynolds number. A mathematical description of the system consists of a mass balance equation that accounts for convection and diffusion, coupled with a bi-Langmuir kinetic equation for the adsorption of the protein on the membrane surface. The assumption of cylindrical membrane pores, uniform radial concentration profile and uniform velocity profile are in order [5].

Boundary layer mass transfer resistance from the bulk liquid phase to the membrane surface can be neglected as it can be easily verified from a time scale analysis of the phenomena involved. The time that a protein needs to diffuse from the centre of the pore to the pore wall is 4 to 5 orders of magnitude lower than the convection time:

$$\tau_d \left(= \frac{d_p^2}{4D} \right) \ll \tau_c \left(= \frac{L}{v} \right) \quad (16)$$

where d_p is the pore diameter, D is the diffusion coefficient of IgG in aqueous solutions, L is the height of the membrane stack and v is the interstitial velocity of the protein solution flowing through the membrane stack. In particular, for the system under investigation $\tau_d \approx 1.3 \times 10^{-3}$ s, while τ_c ranges from 52 to 104 s, for flow rates ranging from 0.5 to 1 ml/min.

The mass balance equation is written over the membrane stack as follows:

$$\varepsilon \frac{\partial c}{\partial t} + \varepsilon v \frac{\partial c}{\partial z} = \varepsilon D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon) \frac{\partial c_{s,1}}{\partial t} - (1 - \varepsilon) \frac{\partial c_{s,2}}{\partial t} \quad (17)$$

and the kinetic equations:

$$\frac{\partial c_{s,i}}{\partial t} = k_{a,i} c(c_{l,i} - c_{s,i}) - k_{d,i} c_{s,i} \quad i = 1, 2 \quad (18)$$

Table 1
Dimensionless numbers used in the mathematical model

$\tau = \frac{tv}{L}$	Dimensionless time
$\zeta = \frac{z}{L}$	Dimensionless spatial variable
$C = \frac{c}{c_0}$	Dimensionless protein concentration in the fluid phase
$C_{in} = \frac{c_{in}}{c_0}$	Dimensionless inlet protein concentration in the fluid phase
$C_{s,i} = \frac{C_{s,i}}{C_{s,i}^0} \quad i = 1, 2$	Dimensionless concentration of protein–ligand complex in the solid phase
$Pe = \frac{vL}{D}$	Axial Péclet number
$m_i = \frac{(1-\varepsilon)c_{l,i}}{\varepsilon c_0} \quad i = 1, 2$	Ratio of total ligand capacity to feed protein concentration
$n_i = \frac{(1-\varepsilon)c_{l,i}k_{a,i}L}{\varepsilon v} \quad i = 1, 2$	Dimensionless number of transfer unit
$r_i = 1 + \frac{\varepsilon v}{K_{d,i}} \quad i = 1, 2$	Dimensionless separation factor

in which k_a and k_d are the kinetic constants of adsorption and desorption, respectively, while the other symbols are illustrated in Fig. 2.

At the beginning of the process, there is no protein solution into the membrane and no protein adsorbed:

$$c = 0 \quad \text{at} \quad z \geq 0, \quad t = 0 \quad (19)$$

$$c_{s,i} = 0 \quad \text{at} \quad z \geq 0, \quad t = 0 \quad i = 1, 2 \quad (20)$$

Danckwerts boundary conditions for frontal analysis were used to account for axial diffusion at the front surface of the membrane, and mixing at the exit of the membrane [21]:

$$\varepsilon v c - \varepsilon D \frac{\partial c}{\partial z} = \varepsilon v c_{in} \quad \text{at} \quad z = 0, \quad t > 0 \quad (21)$$

$$\frac{\partial c}{\partial z} = 0 \quad \text{at} \quad z = L, \quad t > 0 \quad (22)$$

It is worth noticing that the inlet concentration in the membrane stack, c_{in} , does not correspond to the feed protein concentration, c_0 , but to the concentration at the exit of the PFR used to model system dispersion, as described in the previous section.

The problem has been rewritten in dimensionless form by using the dimensionless numbers reported in Table 1. After substitution of the relevant dimensionless numbers the problem formulation becomes:

$$\frac{\partial C}{\partial \tau} + \frac{\partial C}{\partial \zeta} = \frac{1}{Pe} \frac{\partial^2 C}{\partial \zeta^2} - m_1 \frac{\partial C_{s,1}}{\partial \tau} - m_2 \frac{\partial C_{s,2}}{\partial \tau} \quad (23)$$

$$\frac{\partial C_{s,i}}{\partial \tau} = \frac{n_i}{m_i} C(1 - C_{s,i}) - \frac{n_i}{m_i(r_i - 1)} C_{s,i} \quad i = 1, 2 \quad (24)$$

Initial conditions:

$$C = 0 \quad \text{at} \quad \zeta \geq 0, \quad \tau = 0 \quad (25)$$

$$C_{s,i} = 0 \quad \text{at} \quad \zeta \geq 0, \quad \tau = 0 \quad i = 1, 2 \quad (26)$$

Boundary conditions:

$$C - \frac{1}{Pe} \frac{\partial C}{\partial \zeta} = C_{in} \quad \text{at} \quad \zeta = 0, \quad \tau > 0 \quad (27)$$

$$\frac{\partial C}{\partial \zeta} = 0 \quad \text{at} \quad \zeta = 1, \quad \tau > 0 \quad (28)$$

The governing equations are a set of mixed differential equations, one partial differential equation, Eq. (23), and two non-linear ordinary differential equations, Eqs. (24), that must be solved simultaneously.

3. Experimental

A2P-Sartoeoxy affinity membranes have been chosen as a model system and tested in dynamic experiments with pure polyclonal IgG solutions. These membranes have been extensively characterized in batch and dynamic experiments in previous works [15,16].

3.1. Materials and methods

Regenerated cellulose membranes with active epoxy groups, Sartobind epoxy (Sartorius, Göttingen, Germany), have been used as supports for ligand coupling. These reinforced stabilized cellulose membranes have a porosity of about 64%, a nominal pore size of 0.45 μm , an average thickness of 275 μm and a permeability to water of 40–60 ml/(cm² bar min).

The affinity ligand used in this work, mimetic A2P (Prometic Biosciences, Cambridge, UK), is a synthetic ligand obtained by screening of combinatorial ligand libraries [22]. It has been designed to mimic the structure of two key amino acid side chains of Protein A, Phe 132 and Tyr 133, that bind the Fc portion of immunoglobulins [23].

A2P ligand has been coupled onto Sartobind epoxy membranes by Prometic Biosciences, obtaining A2P-Sartoeoxy affinity membranes.

Gammanorm, from Octapharma (Stockholm, Sweden), a polyclonal antibody containing all the four subtypes of human IgG, has been chosen as a source for human IgG. A value of $3.89 \times 10^{-7} \text{ cm}^2/\text{s}$ for the diffusion coefficient of IgG in aqueous solutions has been obtained from literature data [24], and used in all the calculations. Indeed in all cases the Pe number is always larger than 2500 and thus the diffusion term in Eq. (23)

is negligible and use of the axial dispersion instead of molecular diffusivity is not important as it was shown also by Suen and Etzel [5].

Phosphate saline buffer, PBS, at pH 7.4 has been used to dilute the IgG solution in the loading step, and was also used as washing buffer; 0.1 M glycine pH 2.8 has been used as elution buffer. A caustic solution of 1 M NaOH has been used for membrane regeneration.

Flow experiments have been performed using a fast protein liquid chromatography (FPLC) system, Äkta Purifier 100 controlled by Unicorn software, from GE Healthcare (Milan, Italy). Prior to use, the membranes were thoroughly washed with PBS buffer. The membrane layers were then allocated in a membrane holder, 2.5 cm in diameter, connected to the system. The breakthrough curve was measured on-line through UV absorption at 280 nm. All experiments were conducted at room temperature.

3.2. System dispersion curves

System dispersion curves have a fundamental role in order to understand and describe flow non-idealities of the equipment used to perform the dynamic experiments. A system dispersion curve represents the result of a dynamic experiment in non-adsorbing conditions. Hence, in order to remove all the possible interactions between the protein and the membrane matrix, such experiments have been carried out by using Sartobind epoxy membranes without ligand, endcapped with monoethanolamine in order to block the epoxy active groups.

The experiments have been performed using a stack of 10 membrane layers and feeding a dilute IgG solution in PBS buffer at two different values of IgG concentrations, 0.25 and 0.5 mg/ml, and at two different flow rates, 0.5 and 1.0 ml/min.

3.3. Breakthrough curves

The experimental adsorption breakthrough curves have been adopted from a previous work [16]. All the experiments have been performed with a stack of 10 A2P-Sartoepoxy membrane layers at the same experimental conditions used for the system dispersion curves.

At the beginning of the experiments both the system and the membranes were equilibrated by flushing PBS buffer. Then pure IgG solution was fed into the column until complete breakthrough, that is when the membranes were completely saturated. After loading, PBS buffer was used in the washing step, to remove the IgG that was non-specifically and reversibly bound. The washing step continued until the absorbance signal reached the original baseline. A solution of 0.1 M glycine pH 2.8 was used as elution buffer to recover the immobilized IgG. The membranes have been regenerated in batch with 1 M NaOH for 45 min, after the 2nd or after the 3rd cycle.

4. Model solution and parameter estimation

The total volume of the system has been calculated by applying the moment method to the system dispersion curves [25]. By considering a step increase of concentration, the total volume of

the system can be expressed as:

$$V_{\text{sys}} = Q\bar{t} = Q \frac{\int_0^\infty t \frac{dc}{dt} dt}{\int_0^\infty \frac{dc}{dt} dt} = Q \frac{M_1}{M_0} \quad (29)$$

where M_0 and M_1 are the moments of the effluent concentration of zero and first order, respectively [26]. With the value of V_{sys} obtained, the experimental system dispersion curves have been fitted by applying Eq. (15) coupled to Eq. (5), using V_{PFR} as the only fitting parameter.

The set of equations describing the membrane stack behaviour, Eqs. (23)–(28), has been implemented in Comsol Multiphysics 3.2, using the finite element method (FEM).

To illustrate the choice of the adjustable parameters used to best-fit the model to the experimental behaviour of the membrane stack, some considerations are in order. It is not uncommon in affinity chromatography that, for a given system, the static binding capacity and the dynamic binding capacity have different values [27], as it occurs for the model system considered in this work. To account for that aspect, the parameters α_i have been introduced, defined as the ratio between the specific binding capacity, α_1 , (non-specific binding capacity, α_2) and the total binding capacity. In symbols:

$$\alpha_i = \frac{c_{1,i}}{c_1} \quad i = 1, 2 \quad (30)$$

The bi-Langmuir isotherm under dynamic conditions has then been rewritten in the following form:

$$\text{DBC} = \frac{c_1 \alpha_1 c}{K_{d,1} + c} + \frac{c_1 \alpha_2 c}{K_{d,2} + c} \quad (31)$$

DBC is the value of dynamic binding capacity, which is obtained in flow experiments, while the values of $c_{1,i}$ and of $K_{d,i}$ have been determined by fitting batch equilibrium data, and are reported in Table 2. Using Eq. (31) the maximum binding capacity, c_1 , can then be calculated.

In order to compare model calculations and dynamic experimental results, only the two binding kinetic constants, $k_{a,i}$, have been taken as fitting parameters for model validation.

A2P-Sartoepoxy membranes experience a significant decrease in capacity between two subsequent cycles, as already reported in a previous work [16], therefore a different procedure has been applied to simulate the dynamic experiments, depending on the cycle number. The underlining rationale is that with washing and elution steps we do not recover all the protein which was bound on the surface, either specifically or non-specifically, consistently with the fact that the two respective dissociation constants are of the same order of magnitude (Table 2). This implies that in subsequent cycles the ratios α_i are not the same as in the first cycle. Thus, for the

Table 2
Parameters of the bi-Langmuir isotherm obtained by fitting the experimental equilibrium data

$c_{1,1}$ (mg/ml)	2.49
$K_{d,1}$ (mg/ml)	0.036
$c_{1,2}$ (mg/ml)	2.91
$K_{d,2}$ (mg/ml)	0.078

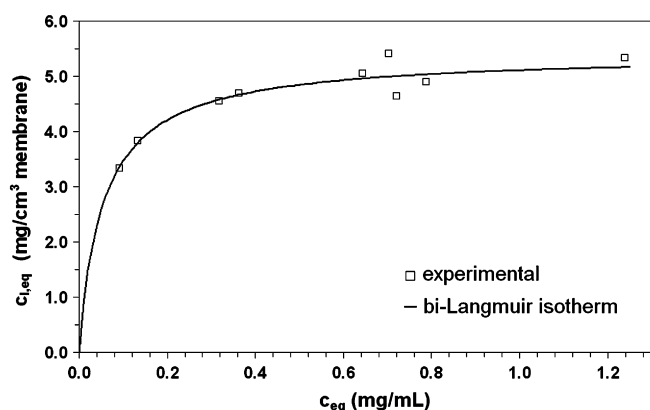


Fig. 3. Equilibrium adsorption isotherm of human IgG onto A2P-Sartopoxy affinity membranes.

cycles following the first one, the specific binding capacity is considered a parameter to best fit the breakthrough curve, the non-specific binding capacity is obtained from Eq. (31), and the kinetic constants, $k_{a,i}$, are given the same values obtained for the first cycle. Therefore, for all the cycles following the first one the only fitting parameter used for model validation is the specific binding capacity of that cycle.

5. Results and discussion

5.1. Batch equilibrium experiments

A2P-Sartopoxy membranes have been fully characterized in batch experiments in terms of binding kinetics, binding capacity, non-specific binding and ageing effects due to repeated use [15]. However, it is important to note that the affinity membranes used are endowed with high values of non-specific binding, even though the membrane support *per se* does not show any non-specific binding capacity. In particular, for HSA a value of about $30 \mu\text{g}/\text{cm}^2$ has been measured [15] and used as input value for the non-specific binding capacity in the simulations.

The adsorption isotherm of human IgG onto A2P-Sartopoxy membranes is reported in Fig. 3. The experimental data have been interpolated by the bi-Langmuir model and the relevant parameters are shown in Table 2. The total static binding capacity is $5.4 \text{ mg}/\text{cm}^3$ solid membrane.

5.2. Dynamic experiments

Using the total system volume obtained from the moment method, the system dispersion curves have been calculated and compared with experimental data. A typical solution is shown in Fig. 4, whilst the volumes of CSTR and PFR contributions are reported in Table 3 for all the experiments performed.

Broadening of the system dispersion curve indicates that flow non-idealities are effectively present in the system under investigation. These effects are well described by the system dispersion model. However, a small discrepancy between the experimental and calculated curves can be noticed in the initial part of the curve; that is due to the fact that the system disper-

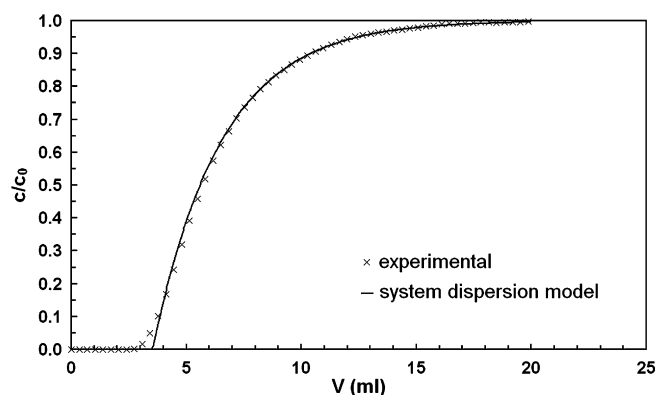


Fig. 4. Comparison between the experimental and calculated system dispersion curves: dimensionless concentration reported versus feed volume.

sion model considered is very simple and uses only one fitting parameter. A more satisfactory fitting of the first portion of the experimental system dispersion curve can be achieved by adding another CSTR in series, after the PFR and the CSTR present in the model, to represent a sigmoid curve with time delay. Nevertheless, with such a new configuration, an additional fitting parameter is required and that is not justified since the residual error is already below 10% with the simple model utilized in this work.

The concentration differences in the feed solution do not influence the shape of dimensionless system dispersion curves, since the system response does not depend on c_0 , as it is evident from Eq. (15). Furthermore, the system dispersion curves are independent of the flow rate for the range of values inspected. This indicates that the regions of the system characterized by plug flow and mixing are well defined; nevertheless, further experimentation at higher flow rates is needed to validate this result under a broader range of operating conditions.

Experimental breakthrough curves have been obtained at two different flow rates and two different concentrations for several subsequent cycles. As observed in batch experiments, the performances of A2P-Sartopoxy affinity membranes decrease with use [15], and those results have been confirmed by the data obtained in dynamic tests. After two or three cycles, the membranes have been regenerated with a 1 M NaOH solution, and an improvement in the membrane performances has been obtained with regeneration, but the initial capacity value has never been completely regained.

The experimental breakthrough curves obtained for the first adsorption cycle using new membranes are reported in Fig. 5 for different operating conditions. As expected, at lower feed protein concentrations, the onset of breakthrough is delayed,

Table 3
Volumes obtained with the moment analysis and the system dispersion model

Q (ml/min)	c_0 (mg/ml)	V_{sys} (ml)	V_{PFR} (ml)	V_{CSTR} (ml)
1.0	0.50	6.61	3.72	2.89
0.5	0.50	6.78	3.70	3.08
0.5	0.25	6.50	3.53	2.97
Average		6.63 ± 0.14	3.65 ± 0.10	2.98 ± 0.10

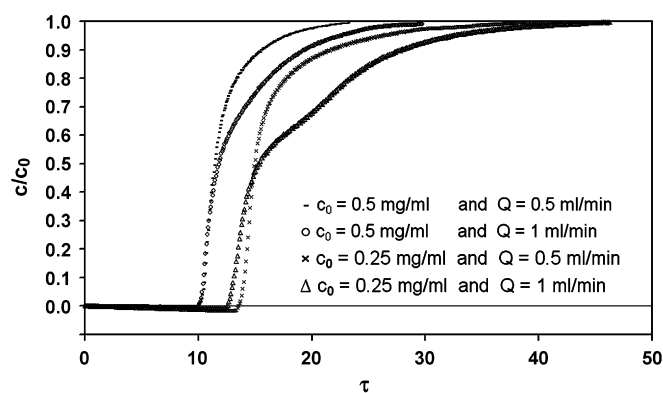


Fig. 5. Comparison among different experimental breakthrough curves for new membranes in the first adsorption cycle: dimensionless concentration reported versus dimensionless time.

since the time required to saturate the membrane is greater. The effects of flow rate on the breakthrough curves are more interesting: all the curves show a sharp initial slope, which decreases with dimensionless time. In the first region of breakthrough the slope of the curve is very similar for all the flow rates and concentrations investigated, while in the second part of the curve the slope depends on the flow rate. This behaviour is consistent with the hypothesis of a bi-Langmuir kinetics and allows a differentiation between the kinetics of the two reactions. The observed trend suggests that a fast reaction occurs at the beginning and is followed by a slower second reaction. The first reaction reaches the same yield for all flow rates inspected, and thus can be considered as an equilibrium reaction. The second reaction does not reach the equilibrium and its yield depends on time; as a consequence, the experimental breakthrough curves are strongly dependent on the flow rate. Particularly, as the flow rate increases, the breakthrough curves become less sharp.

The breakthrough curves calculated with the presented model have been compared with experimental breakthrough curves; the results are reported in Figs. 6–8. The bi-Langmuir model is not only able to describe qualitatively the experimental results, but in most cases is in perfect agreement with the breakthrough curves.

The values of the kinetic constants, $k_{a,1} = 10.01 \times 10^{-3} \pm 0.14 \times 10^{-3} \text{ ml mg}^{-1} \text{ s}^{-1}$ and $k_{a,2} = 0.236 \pm 0.013 \text{ ml mg}^{-1} \text{ s}^{-1}$, obtained as fitting parameters for the first cycle, have been kept constant for the subsequent cycles. As expected, the kinetic constant for non-specific binding is higher than the kinetic constant relative to the specific interaction, confirming that non-specific binding is much faster. An analysis of the characteristic time of reaction for the specific and non-specific binding gives:

$$\tau_{a,1}(c_0 = 0.25 \text{ mg/ml}) = \frac{1}{c_0 k_{a,1}} = 400$$

$$\text{s; } \tau_{a,2}(c_0 = 0.25 \text{ mg/ml}) = \frac{1}{c_0 k_{a,2}} = 17 \text{ s}$$

$$\tau_{a,1}(c_0 = 0.50 \text{ mg/ml}) = \frac{1}{c_0 k_{a,1}} = 200$$

$$\text{s; } \tau_{a,2}(c_0 = 0.50 \text{ mg/ml}) = \frac{1}{c_0 k_{a,2}} = 8.5 \text{ s}$$

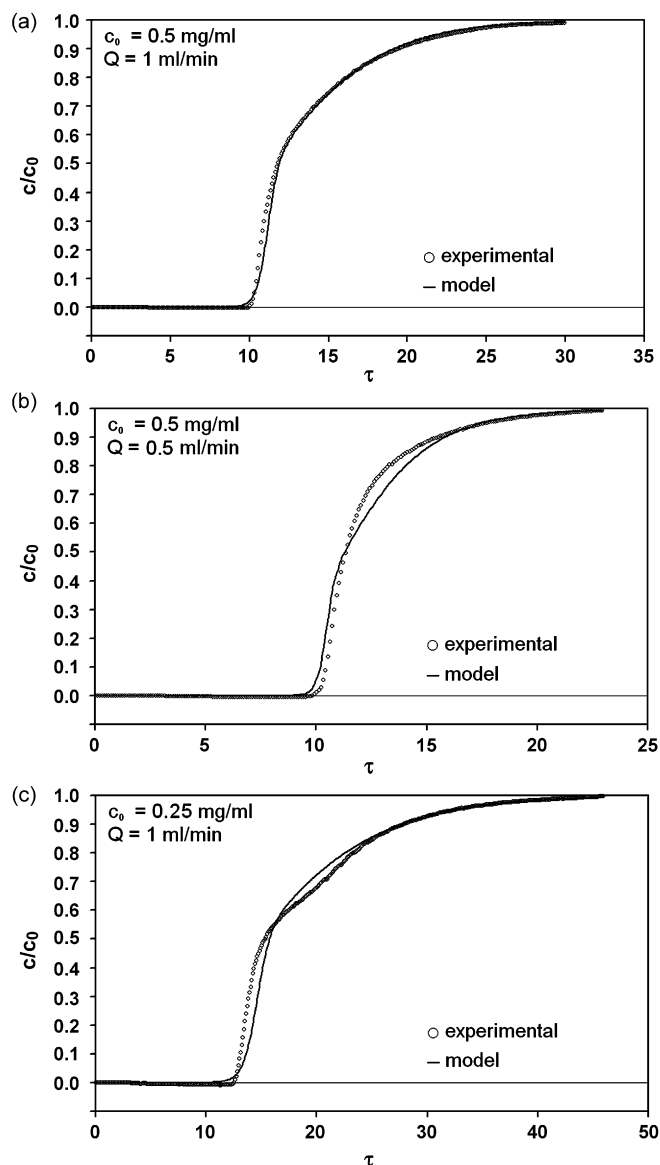


Fig. 6. Comparison between experimental and calculated breakthrough curves at different operating conditions: dimensionless concentration reported versus dimensionless time.

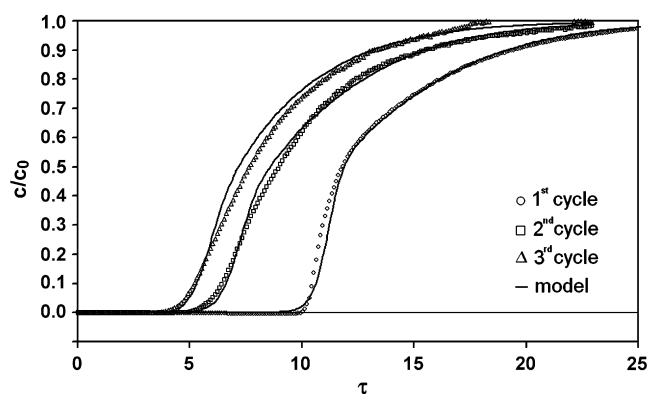


Fig. 7. Comparison between experimental and calculated breakthrough curves for the first three cycles obtained with new membranes at $c_0 = 0.5$ mg/ml and $Q = 1$ ml/min. Dimensionless concentration reported versus dimensionless time.

Table 4

Capacity values obtained from the bi-Langmuir model fitting of the experimental breakthrough curves

	1st cycle	2nd cycle	3rd cycle		1st cycle	2nd cycle	3rd cycle
DBC (mg/ml)	5.15	2.11	0.65	Regeneration	1.04	1.26	0.97
$c_{l,1}$ (mg/ml)	2.65	1.47	0.42		0.96	1.08	0.86
$c_{l,2}$ (mg/ml)	3.09	0.85	0.30		0.17	0.29	0.20
$c_{l,tot}$ (mg/ml)	5.74	2.32	0.72		1.13	1.37	1.06

Operating conditions: $c_0 = 0.5$ mg/ml and $Q = 1$ ml/min.

Table 5

Capacity values obtained from the bi-Langmuir model fitting of the experimental breakthrough curves

	1st cycle	2nd cycle	3rd cycle		1st cycle	2nd cycle	3rd cycle
DBC (mg/ml)	5.17	1.89	1.19	Regeneration	1.62	1.04	1.07
$c_{l,1}$ (mg/ml)	2.89	1.19	0.85		0.97	0.44	0.77
$c_{l,2}$ (mg/ml)	3.37	1.12	0.58		1.01	0.86	0.52
$c_{l,tot}$ (mg/ml)	6.26	2.31	1.43		1.98	1.30	1.29

Operating conditions: $c_0 = 0.25$ mg/ml and $Q = 1$ ml/min.

Table 6

Capacity values obtained from the bi-Langmuir model fitting of the experimental breakthrough curves

	1st cycle	2nd cycle		1st cycle	2nd cycle	3rd cycle
DBC (mg/ml)	4.14	1.57	Regeneration	1.17	0.87	0.69
$c_{l,1}$ (mg/ml)	2.13	1.32		0.63	0.85	0.60
$c_{l,2}$ (mg/ml)	2.48	0.39		0.67	0.09	0.16
$c_{l,tot}$ (mg/ml)	4.61	1.71		1.30	0.94	0.76

Operating conditions: $c_0 = 0.5$ mg/ml and $Q = 0.5$ ml/min.

Table 7

Capacity values obtained from the bi-Langmuir model fitting of the experimental breakthrough curves

	1st cycle	2nd cycle	3rd cycle		1st cycle	2nd cycle
DBC (mg/ml)	4.12	0.84	0.80	Regeneration	0.55	0.91
$c_{l,1}$ (mg/ml)	2.33	0.56	0.90		0.30	0.84
$c_{l,2}$ (mg/ml)	2.71	0.46	0.02		0.38	0.23
$c_{l,tot}$ (mg/ml)	5.04	1.02	0.92		0.68	1.07

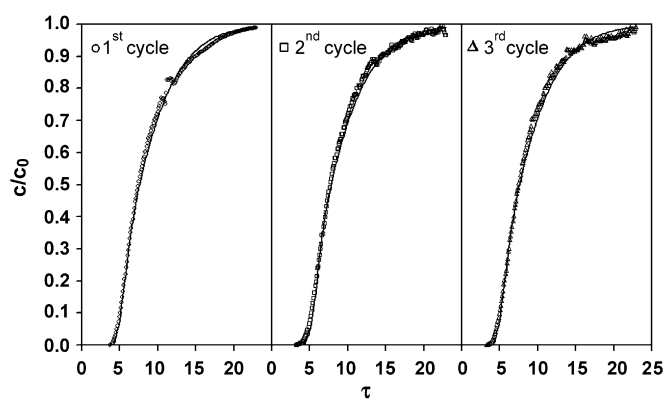
Operating conditions: $c_0 = 0.25$ mg/ml and $Q = 1$ ml/min.

Fig. 8. Comparison between experimental and calculated breakthrough curves for the first three cycles obtained with regenerated membranes at $c_0 = 0.5$ mg/ml and $Q = 1$ ml/min. Dimensionless concentration reported versus dimensionless time.

By reminding that the time scale for convection, τ_c , ranges from 52 to 104 s, one concludes that the time needed for the specific binding reaction is one order of magnitude higher than the time scale for convection, demonstrating that this reaction is not at equilibrium in the range of operating conditions investigated. Instead, the non-specific binding reaction is considerably faster and it is realistic to regard this reaction as at equilibrium.

For every subsequent cycle, where the values of $k_{a,i}$ are the same as in the first cycle, the specific capacity has been taken as fitting parameter and the corresponding values are reported in Tables 4–7.

The dynamic binding capacity decreases with use both for specific and non-specific capacities, as reported in Fig. 9. However, the capacity loss is more pronounced for the non-specific binding capacity and it is more evident going from the first cycle to the second cycle. After regeneration, the original capacity is not totally regained, but it remains almost constant for the subsequent cycles.

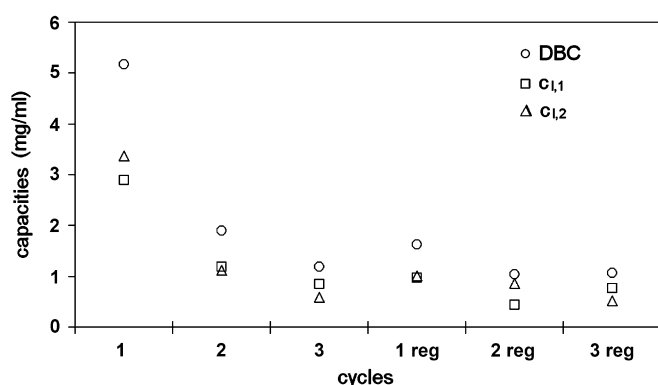


Fig. 9. Trends of the specific, non-specific and total dynamic binding capacity as a function of the cycle number obtained at $c_0 = 0.25$ mg/ml and $Q = 1$ ml/min.

From the data obtained in the first cycle with new membranes the dynamic binding capacity depends on flow rate, as observed by Shiosaki et al., but not on the concentration of the feed solution [28].

6. Conclusions

A mathematical model for the description of protein adsorption on affinity membranes has been proposed and applied to the adsorption of human IgG on A2P-Sartoepoxy affinity membranes.

The model that considers convection, diffusion and bi-Langmuir adsorption has been coupled to a simple system dispersion model to describe the flow non-idealities present in the apparatus under investigation. Despite of its simplicity, the system dispersion model is adequate to describe the system non-idealities.

A good fitting of the experimental data has been obtained with the dynamic model proposed. This model is a useful tool for the description of affinity membrane chromatography behaviour and to predict the membrane performance in such systems, as it is required for process design and process scale up purposes.

For the system investigated, mass transfer is negligible with respect to both convection and reaction. This is a clear indication that the process is limited by the adsorption kinetics between the protein in solution and the ligand immobilized into the matrix, and not by mass transfer diffusion as it happens in bead-based chromatography. Detailed equilibrium experiments are required to find the appropriate binding kinetics which is necessary for a proper description of protein adsorption on affinity membranes.

Nomenclature

A	frontal membrane area (cm^2)
c	protein concentration in the fluid phase (mg ml^{-1})
c_0	feed protein concentration in the fluid phase (mg ml^{-1})
c_l	ligand capacity in the solid phase based on the solid volume (mg ml^{-1})
c_s	concentration of protein–ligand complex in the solid phase (mg ml^{-1})

C	dimensionless protein concentration in the fluid phase
C_s	dimensionless concentration of protein–ligand complex in the solid phase
D	axial diffusion coefficient ($\text{cm}^2 \text{s}^{-1}$)
d_p	pore diameter (cm)
k_a	association rate constant ($\text{ml mg}^{-1} \text{s}^{-1}$)
k_d	dissociation rate constant (s^{-1})
K_d	dissociation equilibrium constant (mg ml^{-1})
L	membrane thickness (cm)
m	ratio of total ligand capacity to feed protein concentration
M_i	effluent concentration moment of i th order
n	dimensionless number of transfer units
Pe	axial Péclet number
Q	flow rate (ml min^{-1})
r	dimensionless separation factor
t	time (s)
$\bar{t}$	average system residence time (s)
t_d	delay time (s)
v	interstitial flow velocity (cm s^{-1})
V	volume (ml)
z	axial distance along the membrane (cm)

Subscripts

1	specific binding
2	non-specific binding
in	inlet
out	outlet
CSTR	variable relative to the CSTR
PFR	variable relative to the PFR
ext	external
sys	system
void	membrane void

Greek letters

α_i	ratio between specific (non-specific) binding capacity and total binding capacity
ε	membrane void fraction
ζ	dimensionless spatial variable
τ	dimensionless time
τ_a	adsorption time scale (s)
τ_c	convection time scale (s)
τ_d	dimensionless delay time
τ_{diff}	diffusion time scale (s)
$\tau_{H'}$	dimensionless time constant

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References

- [1] F. Li, J.X. Zhou, X. Yang, T. Tressel, B. Lee, *BioProcess. J.* 4 (2005) 23.
- [2] D. Low, R. O’Leary, N.S. Pujar, J. *Chromatogr. B* (2006), doi:10.1016/j.jchromb.2006.10.033.

- [3] M.R. Etzel, F. in, T.B. Švec, Z. Tennikova, Deyl (Eds.), *Monolithic Materials* (Journal of Chromatography Library, Vol. 67), Elsevier, Amsterdam, 2003, p. 213.
- [4] K.G. Briefs, M.R. Kula, *Chem. Eng. Sci.* 47 (1992) 141.
- [5] S.Y. Suen, M.R. Etzel, *Chem. Eng. Sci.* 47 (1992) 1355.
- [6] H.C. Liu, J.R. Fried, *AIChE J.* 40 (1994) 40.
- [7] G.C. Serafica, J. Pimbley, G. Belfort, *Biotech. Bioeng.* 43 (1994) 21.
- [8] J.E. Kochan, Y.J. Wu, M.R. Etzel, *Ind. Eng. Chem. Res.* 35 (1996) 1150.
- [9] O.P. Dancette, J.L. Taboureau, E. Tournier, C. Charcosset, P. Blond, *J. Chromatogr. B* 723 (1999) 61.
- [10] H. Yang, M.R. Etzel, *Ind. Eng. Chem. Res.* 42 (2003) 890.
- [11] X. Jin, J. Talbot, G. Tarjus, N.H.L. Wang, *AIChE J.* 40 (1994) 1685.
- [12] I. Lundström, *Prog. Colloid Polym. Sci.* 70 (1985) 76.
- [13] W. Shi, F. Zhang, G. Zhang, *J. Chromatogr. A* 1081 (2005) 156.
- [14] G. Wang, R.G. Carbonell, *J. Chromatogr. A* 1078 (2005) 98.
- [15] C. Boi, G.C. Sarti, *Sep. Sci. Technol.* (submitted for publication).
- [16] C. Boi, S. Dimartino, P. Van Beijeren, G.C. Sarti, *Proceedings of the AIChE Annual Meeting*, American Institute of Chemical Engineers Publ., New York, 2006, p. 240g.
- [17] F.T. Sarfert, M.R. Etzel, *J. Chromatogr. A* 764 (1997) 3.
- [18] S.Y. Suen, *J. Chem. Technol. Biotechnol.* 70 (1997) 278.
- [19] H. Yang, M. Bitzer, M.R. Etzel, *Ind. Eng. Chem. Res.* 38 (1999) 4044.
- [20] S.Y. Suen, M.R. Etzel, *J. Chromatogr. A* 686 (1994) 179.
- [21] P.V. Danckwerts, *Chem. Eng. Sci.* 2 (1953) 1.
- [22] K. Sproule, P. Morrill, J.C. Pearson, S.J. Burton, K.R. Hejnæs, H. Valore, S. Ludvigsen, C.R. Lowe, *J. Chromatogr. B* 740 (2000) 17.
- [23] A.R. Newcombe, C. Cresswell, S. Davies, K. Watson, G. Harris, K. O'Donovan, R. Francis, *J. Chromatogr. B* 814 (2005) 209.
- [24] T. Jøssang, J. Feder, E. Rosenqvist, *J. Protein Chem.* 7 (1998) 165.
- [25] E.N. Lightfoot, A.B. Lenhoff, R.L. Rodriguez, *Chem. Eng. Sci.* 37 (1982) 954.
- [26] D.B. Spalding, *Chem. Eng. Sci.* 9 (1958) 74.
- [27] H. Yang, C. Viera, J. Fischer, M.R. Etzel, *Ind. Eng. Chem. Res.* 41 (2002) 1597.
- [28] A. Shiosaki, M. Goto, T. Hirose, *J. Chromatogr. A* 679 (1994) 1.